# CBSE Class 12 Pe — Sports and Nutrition

## 50 Most-Probable PYQs & Board-Pattern Practice Questions

### 1-Mark (MCQ/Assertion-Reason/VSA)

**Q1.** Which of the following are fat-soluble vitamins? (A) Vitamin D & K (B) Vitamin B & C (C) Vitamin A & E (D) Both A and C [PYQ 2024] | Confidence: Very High

**Q2.** The vitamins that are soluble in water are: (A) Vitamin C and B (B) Vitamin K and E (C) Vitamin D and A (D) All of the above [PYQ] | Confidence: Very High

**Q3.** Given below are two statements labeled Assertion (A) and Reason (R): Assertion (A): The Basal Metabolic Rate (BMR) is the number of calories needed to maintain body function at resting condition. Reason (R): A person who does not engage in any work still requires energy for the functioning of their internal organs. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. [PYQ 2024] | Confidence: Very High

**Q4.** Amino acids and proteins are known as the ______ of life. (A) Building blocks (B) Training blocks (C) Fitness blocks (D) Both A & B [PYQ] | Confidence: High

**Q5.** Given below are two statements labeled Assertion (A) and Reason (R): Assertion (A): The risk of cancer can be reduced by eating more colorful vegetables, fruits, and other plant-foods that have certain phytochemicals. Reason (R): Non-nutritive components of diet are part of a balanced diet. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. [PYQ 2024] | Confidence: High

**Q6.** The BMI index value for an Obese person is: (A) $< 18.5$ (B) $18.5 - 24.9$ (C) $> 30$ (D) $25 - 29.9$ [PYQ] | Confidence: Very High

**Q7.** Which of the following is an example of a macronutrient? (A) Vitamins (B) Minerals (C) Carbohydrates (D) Roughage [Practice] | Confidence: Very High

**Q8.** Scurvy disease is caused due to the lack of: (A) Vitamin A (B) Vitamin B (C) Vitamin C (D) Vitamin D [PYQ] | Confidence: Very High

**Q9.** Goiter is a disease caused due to the deficiency of which mineral? (A) Calcium (B) Iodine (C) Selenium (D) Iron [PYQ] | Confidence: High

**Q10.** Low levels of which specific mineral will lead to Anemia in athletes? (A) Copper (B) Sodium (C) Iron (D) Calcium [PYQ] | Confidence: Very High

**Q11.** Which essential trace mineral is critically responsible for the proper functioning of the thyroid gland? (A) Iron (B) Calcium (C) Iodine (D) Phosphorus [Practice] | Confidence: High

**Q12.** Glucose, Fructose, and Lactose are examples of: (A) Simple Carbohydrates (B) Complex Carbohydrates (C) Minerals (D) Fats [PYQ] | Confidence: High

**Q13.** Vitamin K is primarily needed in the human body for: (A) Healthy bones (B) Cellular metabolism (C) Blood clotting (D) Vision [PYQ] | Confidence: Very High

**Q14.** Assertion (A): Vitamins and minerals are biologically categorized as protective foods. Reason (R): They do not provide energy or calories, but they play a vital role in protecting the body against diseases and maintaining immunity. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. [Practice] | Confidence: Very High

**Q15.** The inability to digest a particular food properly, resulting in stomach ache or nausea, is known as: (A) Food Myths (B) Pitfalls of Dieting (C) Food Intolerance (D) Bulimia [PYQ] | Confidence: Very High

**Q16.** It is recommended to drink ____ of water daily to maintain proper hydration levels. (A) $1-1.5$ litres (B) $2-3$ litres (C) $3.5-4$ litres (D) $0.5-1$ litre [PYQ] | Confidence: Medium

**Q17.** Which of the following is a non-nutritive component of a diet? (A) Fats (B) Proteins (C) Roughage (D) Carbohydrates [PYQ] | Confidence: Very High

**Q18.** Ghee, Butter, Cheese, and Curd are rich sources of ___. (A) Vitamins (B) Fats (C) Minerals (D) Proteins [PYQ] | Confidence: Medium

**Q19.** To calculate an individual's Body Mass Index (BMI), which formula is used? (A) $\frac{\text{Weight (in kg)}}{\text{Height (in meters)}}$ (B) $\frac{\text{Weight (in kg)}}{\text{Height}^2\ (\text{in m}^2)}$ (C) $\frac{\text{Height (in meters)}}{\text{Weight (in kg)}}$ (D) $\frac{\text{Height}^2\ (\text{in m}^2)}{\text{Weight (in kg)}}$ [Practice] | Confidence: Very High

**Q20.** Assertion (A): Consuming high amounts of simple carbohydrates provides a quick and immediate burst of energy to an athlete. Reason (R): Simple carbohydrates have a simple chemical structure and are broken down and digested very rapidly by the body. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. [Practice] | Confidence: High

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## 2-Mark

**Q21.** Differentiate between the nutritive and non-nutritive components of a diet. [PYQ] | Confidence: Very High

**Q22.** Enumerate any two common myths related to food and dieting with their respective facts. [PYQ] | Confidence: Very High

**Q23.** What should be the basic nutrient in a weightlifter's diet and why? [PYQ] | Confidence: High

**Q24.** Define the term 'Food Intolerance' in the context of an athlete's diet. [PYQ] | Confidence: Very High

**Q25.** State the importance of fluid intake during a sports competition. [PYQ] | Confidence: High

**Q26.** Differentiate between Simple Carbohydrates and Complex Carbohydrates by giving one example of each. [PYQ] | Confidence: Very High

**Q27.** Mention any two major "pitfalls of dieting" that an athlete must avoid. [Practice] | Confidence: High

**Q28.** What are Vitamins? Briefly differentiate between fat-soluble and water-soluble vitamins. [PYQ] | Confidence: Very High

**Q29.** Briefly write about minerals as an important nutritive component. Give one example of a macro-mineral and a micro-mineral. [PYQ] | Confidence: Medium

**Q30.** Define Basal Metabolic Rate (BMR). Why is it significant in sports nutrition? [Practice] | Confidence: High

**Q31.** Enlist any four major food sources of proteins in a balanced diet. [Practice] | Confidence: Medium

**Q32.** What is the role of Roughage (fiber) in a sports person's diet? [Practice] | Confidence: High

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### 3-Mark

**Q33.** Mention any three importances of maintaining a proper diet during a sports competition. [PYQ] | Confidence: Very High

**Q34.** "Vitamins are essential for our energy levels and boost the immune system." Comment on this statement. [PYQ] | Confidence: High

**Q35.** Elaborate on the various "pitfalls of dieting" that teenagers often fall into while trying to lose weight rapidly. [PYQ] | Confidence: Very High

**Q36.** Explain the meaning of 'Balanced Diet'. Why is it essential for athletes? [Practice] | Confidence: Very High

**Q37.** Make a list of any three non-nutritive components of diet and briefly explain their roles in the human body. [Practice] | Confidence: High

**Q38.** Discuss any three common food myths prevalent in society regarding weight loss and sports nutrition. [PYQ] | Confidence: High

**Q39.** How is food intolerance treated? What are its common symptoms? Explain in brief. [PYQ 2024] | Confidence: Very High

**Q40.** Differentiate between Macro Nutrients and Micro Nutrients. State the function of any one macro-nutrient. [Practice] | Confidence: Very High

**Q41.** Discuss the physiological importance of Calcium and Iron in an athlete's body. [Practice] | Confidence: Medium

**Q42.** What is Body Mass Index (BMI)? Write the formula to calculate it and list down the World Health Organization (WHO) BMI categories. [Practice] | Confidence: High

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## 5-Mark (Long Answer/Case-Study)

**Q43.** What do you understand by the non-nutritive components of diet? Explain the importance of any two such components (e.g., Water, Roughage, Artificial Sweeteners) in detail. [PYQ] | Confidence: Very High

**Q44.** The dietary requirements of an athlete change drastically before, during, and after an event. Explain in detail the Pre-competition, During-competition, and Post-competition dietary strategies and nutrient requirements for an endurance athlete to ensure peak performance. [Practice] | Confidence: Very High

**Q45.** Write a detailed note on "Food Intolerance". Additionally, critically analyze the major "Pitfalls of Dieting" among modern sports athletes. [PYQ] | Confidence: High

**Q46.** What is a Balanced Diet? Elaborate comprehensively on the functions and specific importance of any four Macro Nutrients (Carbohydrates, Proteins, Fats, Water) in a sportsperson's regular diet. [Practice] | Confidence: Very High

**Q47. Case Study:** A balanced diet refers to the intake of food constituting all the necessary nutrients. Ram shares his knowledge of ‘food and nutrition’ with neighbors while visiting his grandparents in a village. Ram notices that a few people living in that village are suffering from goiter and severe anemia. (i) Minerals are placed under the __ **nutrient category on the basis of required quantity. (ii) Goiter is caused due to the deficiency of** ____. (iii) Low levels of which mineral lead to Anemia? (iv) Name the category of vitamins that are stored in the body's fatty tissues (Fat-soluble or Water-soluble). [PYQ] | Confidence: Very High

**Q48. Case Study:** Shruti, a gymnast, started following a very strict diet plan to lose weight quickly before the national trials. She skipped meals, completely avoided carbohydrates, and took diet pills. Within a week, she started feeling dizzy, lacked explosive energy during her routines, and suffered from muscle cramps. (i) As a physical education student, what do you feel are the pitfalls of this dieting plan? (ii) What will be your scientific advice to Shruti to control her weight properly? (iii) Carbohydrates are divided into two types: simple and ___. (iv) What is the meaning of the term 'Pitfalls of Dieting'? [PYQ] | Confidence: High

**Q49. Case Study:** The Ganga school teams have started practice for the upcoming Basketball Cluster Tournament. One day, the school secretary visited the playground and felt that the players were looking weak and fatigued early in the session. After discussion with the coach, he arranged a dietician to evaluate the players’ dietary requirements. (i) Which type of nutrient is strictly advisable for players for immediate energy recovery? (ii) Proteins and amino acids are commonly referred to as the __ **of life. (iii) It is recommended to drink** __ litres of water daily to prevent dehydration during practice. (iv) Vitamins and minerals are also referred to as _____ foods because they build immunity. [PYQ] | Confidence: Very High

**Q50. Case Study:** Below given is the BMI data of a school's health check-up: 15% are $< 18.5$, 50% are $18.5 - 24.9$, 25% are $25 - 29.9$, and 10% are $> 30$. (i) According to the WHO formula $\frac{\text{Weight}}{\text{Height}^2}$, in which category does the major student population (50%) fall? (ii) The school has to develop an activity-based program to decrease the number of students in the $> 30$ category. What does a BMI $> 30$ signify? (iii) Which specific BMI range category is classified as "Underweight"? (iv) State any one non-nutritive component of a diet that obese students should increase to feel full without adding calories. [PYQ] | Confidence: High

## Answer Key

**Q1.** (D) Both A and C (Vitamins A, D, E, K are fat-soluble). **Q2.** (A) Vitamin C and B. **Q3.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q4.** (A) Building blocks. **Q5.** (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A) (Phytochemicals reduce cancer risk, and non-nutritive components are part of a diet, but R does not explain why A happens). **Q6.** (C) $> 30$. **Q7.** (C) Carbohydrates. **Q8.** (C) Vitamin C. **Q9.** (B) Iodine. **Q10.** (C) Iron. **Q11.** (C) Iodine. **Q12.** (A) Simple Carbohydrates. **Q13.** (C) Blood clotting. **Q14.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q15.** (C) Food Intolerance. **Q16.** (B) $2 - 3$ litres. **Q17.** (C) Roughage. **Q18.** (B) Fats. **Q19.** (B) $\frac{\text{Weight (in kg)}}{\text{Height}^2 \text{ (in m}^2\text{)}}$. **Q20.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q21.** Nutritive components provide calories/energy (e.g., Carbs, Proteins, Fats). Non-nutritive components do not provide calories but are essential for body functions (e.g., Water, Roughage, Artificial Sweeteners). **Q22.** Myth 1: Starve yourself to lose weight. Fact: Starving lowers metabolism and causes weakness. Myth 2: Carbohydrates make you fat. Fact: Excess calories make you fat, complex carbs are essential for energy. **Q23.** Protein. Weightlifters undergo heavy resistance training which causes muscle tissue micro-tears. Proteins are the "building blocks" required for the rapid repair and hypertrophy of these muscle fibers. **Q24.** Food intolerance is the inability of the digestive system to properly break down and digest a certain type of food (e.g., lactose or gluten), leading to gastrointestinal issues like gas, bloating, or stomach ache. **Q25.** Sweating during competition causes fluid and electrolyte loss. Fluid intake prevents dehydration, maintains body temperature (thermoregulation), and prevents muscle cramps and premature fatigue. **Q26.** Simple Carbs: Quick to digest, provide immediate energy (e.g., Glucose, Sugar). Complex Carbs: Take longer to digest, provide sustained energy over time (e.g., Starch, Whole grains). **Q27.** 1. Extreme reduction of calories leading to nutrient deficiency. 2. Skipping meals leading to a drop in metabolic rate and energy levels. **Q28.** Vitamins are organic compounds required in small quantities for immunity and metabolism. Fat-soluble (A, D, E, K) are stored in body fat. Water-soluble (B, C) dissolve in water and are not stored. **Q29.** Minerals are micro-nutrients essential for bone health, fluid balance, and hormone production. Macro-mineral example: Calcium (needed in larger amounts). Micro-mineral example: Iron or Iodine (needed in trace amounts). **Q30.** BMR is the minimum amount of energy (calories) required to maintain vital internal organ functions (like the heart and lungs) at complete rest. It helps athletes calculate their baseline caloric needs before adding training calories. **Q31.** Eggs, Fish, Meat, Pulses/Lentils, Soybeans, Milk. **Q32.** Roughage (fiber) adds bulk to the diet, satisfies appetite without adding calories, prevents constipation, and ensures smooth functioning of the digestive tract. **Q33.** 1. Sustains energy levels (glycogen stores) to delay fatigue. 2. Prevents dehydration and muscle cramps. 3. Maintains blood sugar levels to keep the athlete focused and alert. **Q34.** Yes. Although vitamins do not provide direct calories (energy), they act as catalysts in metabolic reactions that release energy from carbohydrates and fats. They also form antibodies that boost the immune system against diseases. **Q35.** 1. Skipping Meals: Slows down metabolism. 2. Extreme Caloric Restriction: Causes fatigue and severe nutritional deficiencies. 3. Dehydration: Relying on liquid diets or sweating to lose weight damages kidneys and causes cramps. **Q36.** A balanced diet contains adequate amounts of all macro-nutrients (carbs, proteins, fats) and micro-nutrients (vitamins, minerals) along with water and roughage. Essential for athletes to fuel intense training, repair tissues, and optimize recovery. **Q37.** 1. Water: Regulates body temperature and transports nutrients. 2. Roughage: Aids digestion and prevents constipation. 3. Phytochemicals/Color compounds: Act as antioxidants to reduce inflammation. **Q38.** 1. Drinking water while eating makes you fat (False: water has 0 calories). 2. Fat-free products are always healthy (False: they often have high added sugar). 3. Skipping breakfast aids weight loss (False: it drops metabolism and leads to overeating later). **Q39.** Treatment involves identifying the specific trigger food and eliminating or reducing it from the diet, sometimes substituting it with an alternative (e.g., almond milk instead of dairy). Symptoms include nausea, stomach ache, bloating, vomiting, and diarrhea. **Q40.** Macronutrients are needed in large quantities (Carbs, Proteins, Fats) to provide energy and build tissue. Micronutrients are needed in small quantities (Vitamins, Minerals) for disease prevention. Function of Carbs: Primary fuel source for the brain and working muscles. **Q41.** Calcium is essential for bone density, teeth formation, and muscle contraction mechanism. Iron is critical for forming hemoglobin in red blood cells, which transports oxygen to working muscles. **Q42.** BMI is a statistical measure to assess body fat based on height and weight. Formula: $\frac{\text{Weight in kg}}{\text{Height in } m^2}$. Categories: $< 18.5$ (Underweight), $18.5 - 24.9$ (Normal), $25 - 29.9$ (Overweight), $> 30$ (Obese). **Q43.** Non-nutritive components do not provide energy/calories but are vital for survival. 1. Water: Constitutes ~60% of body weight, vital for nutrient transport, waste removal, and thermoregulation (sweating). 2. Roughage: Undigestible plant fiber that provides bulk to food, aids in bowel movements, and prevents digestive disorders. 3. Artificial Sweeteners: Provide sweetness without calories, aiding in weight control. **Q44.** Pre-competition: High in complex carbs (for glycogen stores), moderate protein, low fat/fiber (to avoid stomach upset), consumed 3-4 hours prior. During-competition: Simple carbs (glucose drinks) and electrolytes/fluids to maintain hydration and blood sugar. Post-competition: High protein (for muscle repair), carbs (to replenish glycogen), and plenty of fluids to rehydrate. **Q45.** Food Intolerance means the digestive system lacks the enzymes to break down certain foods, causing gastrointestinal distress without involving the immune system. Pitfalls of Dieting: 1. Starvation/Skipping meals reduces BMR. 2. Restricting entire food groups causes vitamin/mineral deficiencies. 3. Taking unprescribed diet pills causes heart and kidney stress. 4. Rapid weight loss leads to muscle loss instead of fat loss. **Q46.** Balanced diet provides all nutrients in the correct ratio. 1. Carbohydrates: Main energy source (provide $4$ kcal/g); essential for high-intensity work. 2. Proteins: Building blocks ($4$ kcal/g); repair micro-tears in muscles after workouts. 3. Fats: Concentrated energy ($9$ kcal/g); protect internal organs and absorb vitamins A, D, E, K. 4. Water: 0 calories; regulates temperature and hydrates cells. **Q47.** (i) Micro-nutrient. (ii) Iodine. (iii) Iron. (iv) Fat-soluble vitamins. **Q48.** (i) Extreme weight loss, loss of muscle mass, dehydration, and nutrient deficiency. (ii) Eat a balanced diet, maintain a slight caloric deficit, and do regular exercise instead of starving. (iii) Complex carbohydrates. (iv) The negative effects, drawbacks, or dangers of unscientific weight-loss methods. **Q49.** (i) Simple Carbohydrates (Glucose). (ii) Building blocks. (iii) $2 - 3$ litres. (iv) Protective foods. **Q50.** (i) Normal weight. (ii) Obesity (excessive body fat posing severe health risks). (iii) $< 18.5$. (iv) Roughage / Fiber (or Water).

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CBSE Class 12 (2027) — Chapterwise PYQ Bank. Compiled from past board papers + board-pattern practice questions (clearly labelled).